Aditya Raut

San Jose, California | adityaanilraut@gmail.com | +1 (669) 499-7554 | linkedin.com/in/adityaanilraut | github.com/adityaanilraut

Summary

Software Engineer with **3+ years** of experience building production-grade **Python** applications, backend systems, and AI-powered software. Hands-on experience with **LLM applications, semantic search, RAG pipelines, vector databases, model optimization**, and scalable, distributed data processing on **AWS**.

Technical Skills

Languages: Python, Java, Go, C/C++, SQL, JavaScript, TypeScript, Bash
Machine Learning: PyTorch, Hugging Face Transformers, Deep Learning, Neural Networks, Transfer Learning, Fine-tuning, LoRA, PEFT, Quantization, CUDA
LLM Systems: RAG, Embeddings, FAISS, ChromaDB, LangChain, LangGraph, MCP, Vector Search, Multi-Agent Systems
Data Engineering: Pandas, NumPy, PySpark, Databricks, Feature Engineering, SQL Optimization
Backend & Cloud: FastAPI, REST APIs, Docker, Kubernetes, AWS (Lambda, ECS, EKS, EC2, S3, SageMaker), Redis, PostgreSQL, MongoDB, Kafka, GitHub Actions, Terraform

Professional Experience

California State University, Chico — *Research Assistant* Aug 2025 – May 2026

- Conducted comparative performance research on **Rust vs. C++** for CPU and GPU workloads, benchmarking execution speed, memory usage, and parallel throughput across compute-intensive tasks (GitHub).
- Designed and implemented GPU-accelerated benchmarks, analyzing low-level performance trade-offs and safety-versus-speed characteristics between the two languages.

Tata Consultancy Services (TCS) — *System Engineer* Apr 2021 – Oct 2023

Clients: GE Aerospace, Reserve Bank of India

- Developed production backend applications in **Python** and **Java**, building scalable REST APIs and distributed services for mission-critical financial systems serving **1,000+ regulated financial institutions**.
- Designed and deployed an enterprise **semantic search platform** using dense embeddings, vector retrieval, and metadata indexing across **100,000+ documents**, improving search relevance by **35%** and reducing query latency by **40%**.
- Built scalable **Python data pipelines** with **PySpark**, Databricks, and AWS; optimized SQL queries and schemas, cutting reporting latency by **30%** while increasing pipeline throughput.
- Containerized services with Docker, deployed on AWS ECS and Kubernetes, and implemented CI/CD with automated monitoring via CloudWatch and Prometheus.

Launchpad Technology — *Software Engineer* May 2020 – Apr 2021

- Engineered backend microservices in **Python**, **Java**, and **Node.js**, exposing **10+ production REST APIs** for customer-facing applications.
- Deployed cloud-native applications on **AWS**, improving availability to **99.9%** and reducing deployment issues by **25%**; optimized Oracle SQL queries, procedures, and indexes, reducing response time by **25%**.

Technical Projects

TurboQuant: KV-Cache Quantization for LLM Inference (GitHub) *PyTorch, CUDA, Quantization*

- Implemented FP16→INT4 KV-cache quantization in PyTorch, reducing inference memory by ∼**4**× and enabling longer context windows; benchmarked throughput, latency, GPU memory, and accuracy across quantization strategies.

AI Context Router (GitHub) *FastAPI, Python, Docker, LLMs*

- Built an intelligent routing platform that dynamically selects LLM providers by reasoning complexity, latency, token usage, and cost; developed async FastAPI services with Dockerized deployment and health monitoring.

CoderAI: Multi-Agent Code Intelligence CLI (GitHub) *Python, MCP, Multi-Agent Systems*

- Built a multi-agent engineering assistant (planner, coder, reviewer agents) for repository analysis, code generation, debugging, and iterative repair, with sandboxed execution and tool calling via the Model Context Protocol (MCP).

Semantic Search Engine (GitHub) *Python, RAG, FAISS, Sentence Transformers*

- Developed an end-to-end RAG pipeline with document ingestion, embedding generation, metadata filtering, reranking, and FastAPI inference; built scalable FAISS indexes, improving Top-3 retrieval precision over baseline vector search.

Education

California State University, Chico — M.S. Computer Science Aug 2024 – May 2026
University of Mumbai — B.E. Computer Engineering Aug 2017 – May 2020

Achievements

Wefunder AI Hackathon Winner — Built an intelligent LLM routing platform selecting optimal models by complexity, latency, tokens, and cost.
TCS Awards — Best Team Award (2022), On the Spot Team Award (2022), On the Spot Award (2023)